

Automated Rice Storage and Thermal Disinfestation System: An IoT-Based Approach for Post-Harvest Pest Management

Post-harvest losses from biological agents and environmental instability threaten food security and farm incomes. This document describes "Rice Saver" — a microcontroller-driven, chemical-free system that combines continuous micro-climate control, motion sensing, and a scheduled thermal eradication cycle at 60°C to eliminate *Sitophilus oryzae* across developmental stages while preserving grain quality.

SILO 7
KEEP CLEAR
AUTHORIZED
PERSONNEL ONLY

DANGER
HIGH TEMPERATURE
KEEP OUT

Introduction

Grain storage exposes rice to abiotic stresses (temperature, moisture) and biotic threats (insects, fungi, rodents). Conventional fumigation carries ecological and health risks and faces regulatory and resistance challenges. Rice Saver offers an automated, residue-free alternative that reduces reliance on chemicals and aligns with residue-sensitive supply chains.



Target Pest: *Sitophilus oryzae*

Primary grain feeder whose larvae develop within kernels, making early detection difficult. Thrives 27–30°C and 60–80% RH.



Abiotic Risks

Elevated RH and fluctuating temperature promote fungal growth and lipid oxidation; both reduce market value and safety.

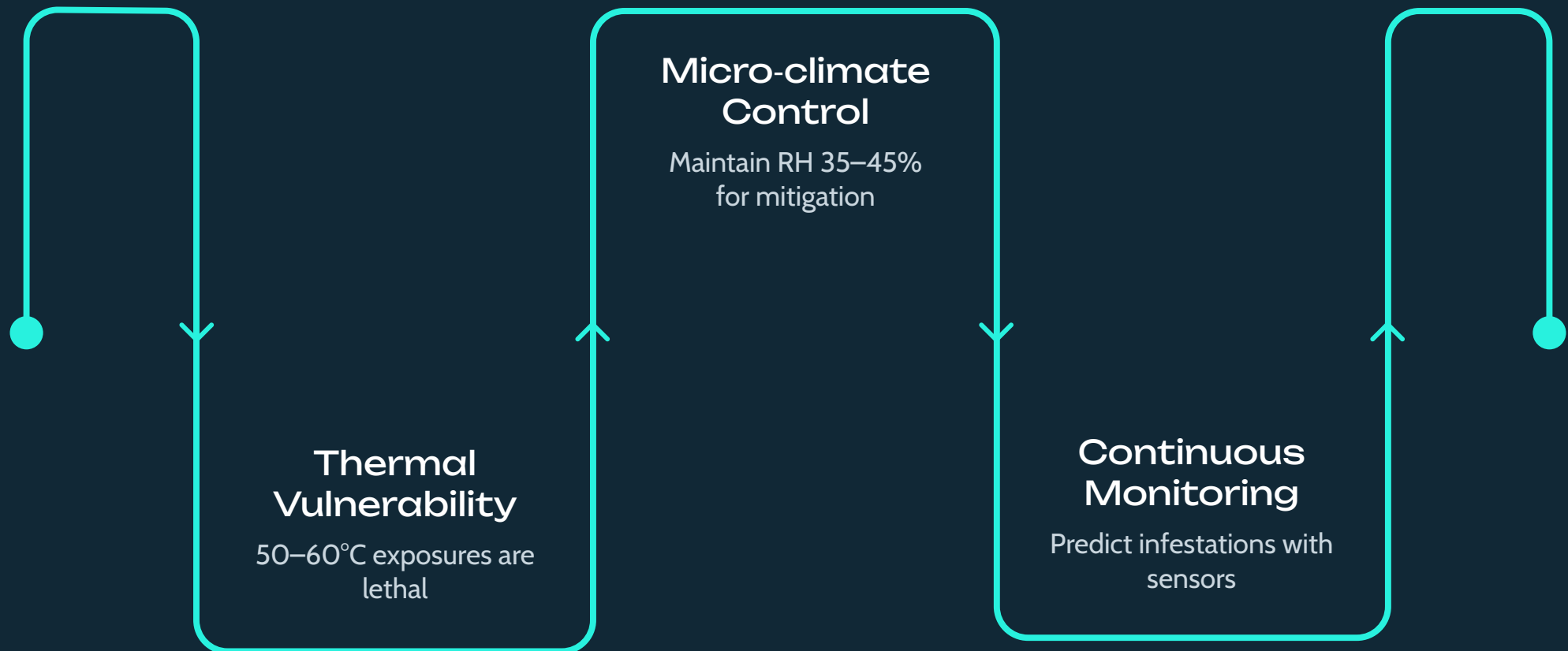


System Objectives

Maintain safe micro-climate, preserve nutrition, and perform automated thermal pest eradication without chemical residues.

Literature Review — Thermal Vulnerabilities & Micro-climate

Thermal interventions and environmental control are well-supported: cold storage below 10°C reduces survival, while short exposures above 50°C rapidly increase mortality. Empirical studies report 60°C exposures achieving near-complete mortality of *S. oryzae* across life stages with negligible changes to moisture and key quality metrics when ventilation and post-treatment conditioning are properly managed.



Continuous sensing of temperature and RH yields leading indicators of pest population dynamics. Integrating those data into automated actuators allows pre-emptive action before infestations reach economically damaging thresholds.

System Architecture & Hardware Design

Rice Saver is engineered for practical deployment: an Arduino-class microcontroller governs sensor polling, logic rules, and actuator control; relay modules mediate high-voltage heating elements and exhaust fans; and a local 16x2 I2C LCD provides operational telemetry. Design prioritises electrical isolation, dust resilience, and field-serviceability.



Control Unit

Arduino or equivalent MCU with watchdog and realtime scheduling; firmware implements safety interlocks and thermal control loops.



Sensor Suite

DHT22 (temperature & RH) probes embedded at multiple depths; PIR sensors around perimeters for macro-pest detection.



Actuation

Solid relay arrays and contactors control heating elements and exhaust fans; current sensing provides fault detection.



Local UI & Alarms

16x2 I2C LCD and multicolour LEDs display states: Safe, Warning, Thermal Cycle Active, and Intrusion Count.

Methodology & Operational Logic

Operation comprises three autonomous routines: baseline micro-climate control, scheduled thermal eradication, and independent motion intrusion logging. Control logic enforces safety bounds, sequences heating with forced ventilation, and records events for traceability.

O1

1. Baseline Monitoring

Maintain RH 35–45% and temperature 17–22°C. Exceeding 45% RH triggers immediate ventilation and an alarm state to prevent fungal establishment.

O2

2. Thermal Eradication

Manual or scheduled cycle raises internal temperature to 60°C for two hours with controlled airflow to ensure heat penetration, followed by staged venting back to 20°C to protect rice lipids.

O3

3. Intrusion Tracking

PIR interrupts increment a Motion Count logged locally; time-stamped events help quantify rodent pressure and inform physical exclusion improvements.

Validation, Safety & Quality Assurance

Thermal treatment validation requires instrumented trials with probe temperatures at multiple grain depths and moisture analyses pre/post-treatment. Safety systems include thermal cutouts, overcurrent protection, and interlocks preventing heating if ventilation is compromised. QA protocols track motion counts, cycle durations, and sensor calibrations to maintain regulatory compliance and product traceability.



Recommended QA Tests

- Multi-point temperature profiling during thermal cycles
- Pre/post moisture and free fatty acid assays
- Sensor calibration schedule and relay actuation logs

Record logs locally and optionally stream to the cloud for audit and analysis.

Scalability & Future Integration

The modular control logic supports horizontal scaling across facility silos. Future development includes MQTT telemetry, cloud ingestion (e.g., FastAPI/Spring Boot backends), and analytics for seasonal outbreak prediction and centralised command of eradication cycles. Field trials should explore adaptive scheduling driven by predictive models built from time-series sensor data.



Conclusions & Recommendations

Rice Saver demonstrates a practical, non-chemical strategy for reducing post-harvest losses from *Sitophilus oryzae*, fungal growth, and rodent intrusion. Key recommendations for deployment: instrumented validation of thermal cycles, staged ventilation protocols to protect grain chemistry, and phased roll-out with cloud telemetry for aggregated analytics and remote control.

Immediate Steps

Run instrumented pilot on a sample silo; validate thermal uniformity and quality retention.

Mid-Term

Integrate MQTT telemetry and central dashboard; implement predictive maintenance for sensors and relays.

Long-Term

Refine algorithms for adaptive scheduling and explore energy optimisation (waste heat recovery).

References and experimental protocols are included to support adoption by agricultural engineers and post-harvest researchers.